

Calculus — Quiz 2

Show all work clearly. Answers without justification receive no credit. Calculators and reference/formula sheets are NOT permitted.

1. (8%) Coordinate Systems

- (a) (3%) Convert the cylindrical point $(r, \theta, z) = (4, \frac{5\pi}{6}, -2)$ to rectangular coordinates.
- (b) (3%) Convert the rectangular point $(x, y, z) = (\sqrt{3}, -1, 2)$ to cylindrical and spherical coordinates.
- (c) (2%) Convert the spherical point $(\rho, \theta, \phi) = (6, \frac{\pi}{3}, \frac{\pi}{6})$ to cylindrical and rectangular coordinates.

2. (12%) Let

$$\mathbf{r}(t) = \langle t^2, e^t, \ln(t+1) \rangle.$$

- (a) (4%) Evaluate $\lim_{t \rightarrow 0^+} \mathbf{r}(t)$.
- (b) (4%) Find the domain of $\mathbf{r}$ and the interval(s) on which $\mathbf{r}$ is smooth.
- (c) (4%) Let $\mathbf{g}(t) = \langle e^{-t}, t, t^2 \rangle$. Compute

$$\left. \frac{d}{dt} (\mathbf{r}(t) \cdot \mathbf{g}(t)) \right|_{t=1}.$$

3. (12%) Evaluate the following limits, or show that the limit does not exist.

- (a) (6%) $\lim_{(x,y) \rightarrow (0,0)} \frac{\sqrt{4+x^2+y^2} - 2}{x^2 + y^2}$
- (b) (6%) $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \sin\left(\frac{1}{x^2 + y^2}\right)$

4. (18%)

- (a) (6%) Define

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Compute $f_x(0, 0)$ and $f_y(0, 0)$. Then decide whether f is differentiable at $(0, 0)$.

- (b) (6%) Let $z = f(x, y) = xe^y$, where

$$x = u^2 + v, \quad y = u - v^2.$$

Find z_u and z_v at $(u, v) = (1, 1)$.

- (c) (6%) Find the tangent plane to the surface

$$x^2 - y^2 + z^2 = 1$$

at the point $(1, 1, 1)$.

5. (6%) Let $f(x, y) = \ln(1 + x^2 + y^2)$. At $P = (1, 2)$, find the directional derivative in the direction from P to $Q = (4, 6)$. Also find the direction and rate of steepest increase.

6. (8%) Find and classify all critical points of

$$f(x, y) = x^3 - 3x + y^2 - 4y.$$

7. (16%) Evaluate the following double integrals.

- (a) (5%) Let R be the region

$$0 \leq x \leq 1, \quad x \leq y \leq 1.$$

Reverse the order of integration and evaluate

$$\int_0^1 \int_x^1 2y \, dy \, dx.$$

- (b) (6%) Let D be the region between $y = x^2$ and $y = 2x$. Sketch or describe D , and evaluate

$$\iint_D x \, dA.$$

- (c) (5%) Use polar coordinates to evaluate

$$\iint_E (x^2 + y^2) \, dA,$$

where E is the region in the first quadrant between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

8. (6%) Set up and evaluate the surface area of

$$z = 2x + 3y$$

over the rectangle $0 \leq x \leq 1, 0 \leq y \leq 2$.

9. (6%) Use a double integral to find the volume of the solid under the surface

$$z = 6 - x - 2y$$

and above the rectangle $0 \leq x \leq 1, 0 \leq y \leq 2$.

10. (8%) Use cylindrical coordinates to set up and evaluate a triple integral for the volume of the solid below the paraboloid

$$z = 9 - r^2$$

and above the xy -plane.